

```
mysql> CREATE DATABASE bookflow_db;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> USE bookflow_db;
Database changed
```

```
mysql> CREATE TABLE Books (
  -> book_id INT PRIMARY KEY,
  -> title VARCHAR(100) NOT NULL,
  -> isbn VARCHAR(20) UNIQUE,
  -> published_year INT CHECK (published_year < 2027)
  -> );
```

```
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES
  -> (1, 'The Great Gatsby', '9780743273565', 1925),
  -> (2, 'To Kill a Mockingbird', '9780061120084', 1960),
  -> (3, '1984', '9780451524935', 1949);
```

```
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Books;
```

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949

```
3 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Members (
  -> member_id INT PRIMARY KEY,
  -> full_name VARCHAR(100),
  -> email VARCHAR(100) UNIQUE
  -> );
```

```
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> INSERT INTO Members (member_id, full_name, email) VALUES
  -> (101, 'John Smith', 'john.smith@email.com'),
  -> (102, 'Emma Wilson', 'emma.wilson@email.com'),
  -> (103, 'Michael Brown', 'michael.brown@email.com');
```

```
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Members;
```

member_id	full_name	email
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

```
3 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Loans (  
  -> loan_id INT PRIMARY KEY,  
  -> member_id INT,  
  -> book_id INT,  
  -> loan_date DATE,  
  -> FOREIGN KEY (member_id) REFERENCES Members(member_id),  
  -> FOREIGN KEY (book_id) REFERENCES Books(book_id)  
  -> );
```

```
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),  
  -> (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),  
  -> (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),  
  -> (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),  
  -> (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check  
, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),  
(3, 103, 3, '2025-01-10')' at line 1
```

```
mysql> SELECT * FROM Loans;
```

```
Empty set (0.00 sec)
```

```
mysql> SELECT m.full_name AS Member_Name, b.title AS Book_Title  
  -> FROM Loans l  
  -> INNER JOIN Members m ON l.member_id = m.member_id  
  -> INNER JOIN Books b ON l.book_id = b.book_id;
```

```
Empty set (0.00 sec)
```

```
mysql>
```

```
mysql> SELECT published_year, COUNT(book_id) AS Total_Books  
  -> FROM Books  
  -> GROUP BY published_year  
  -> ORDER BY published_year;
```

```
+-----+
```

published_year	Total_Books
1925	1
1949	1
1960	1

3 rows in set (0.01 sec)

```
mysql> CREATE TABLE Donation_History (
->     donation_id INT PRIMARY KEY,
->     book_id INT,
->     donor_name VARCHAR(100),
->     donation_date DATE,
->     FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
```

Query OK, 0 rows affected (0.03 sec)

```
mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> INSERT INTO Books (book_id, title, isbn, published_year)
-> VALUES (4, 'Animal Farm', '9780451526342', 1945);
Query OK, 1 row affected (0.00 sec)
```

```
mysql>
mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
-> VALUES (1, 4, 'Raj Kumar', CURDATE());
Query OK, 1 row affected (0.00 sec)
```

```
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> CREATE INDEX idx_books_isbn ON Books(isbn);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Books WHERE isbn = '9780451524935';
+-----+-----+-----+-----+
| book_id | title | isbn          | published_year |
+-----+-----+-----+-----+
|      3 | 1984  | 9780451524935 |      1949      |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```